



PHYSICS LABORATORY

EXPERIMENT 5.

MAGNETIC FIELD OF SINGLE COILS/ BIOT-SAVART LAW

Content



1. Objectives
2. Equipments
3. Theory
4. Experiments
5. Report



Objectives

- The magnetic field along the axis of wire loops and coils of different dimensions is measured with a teslameter (Hall probe).
- The relationship between the maximum field strength and the dimensions is investigated and a comparison is made between the measured and the theoretical effects of position.

Equipments

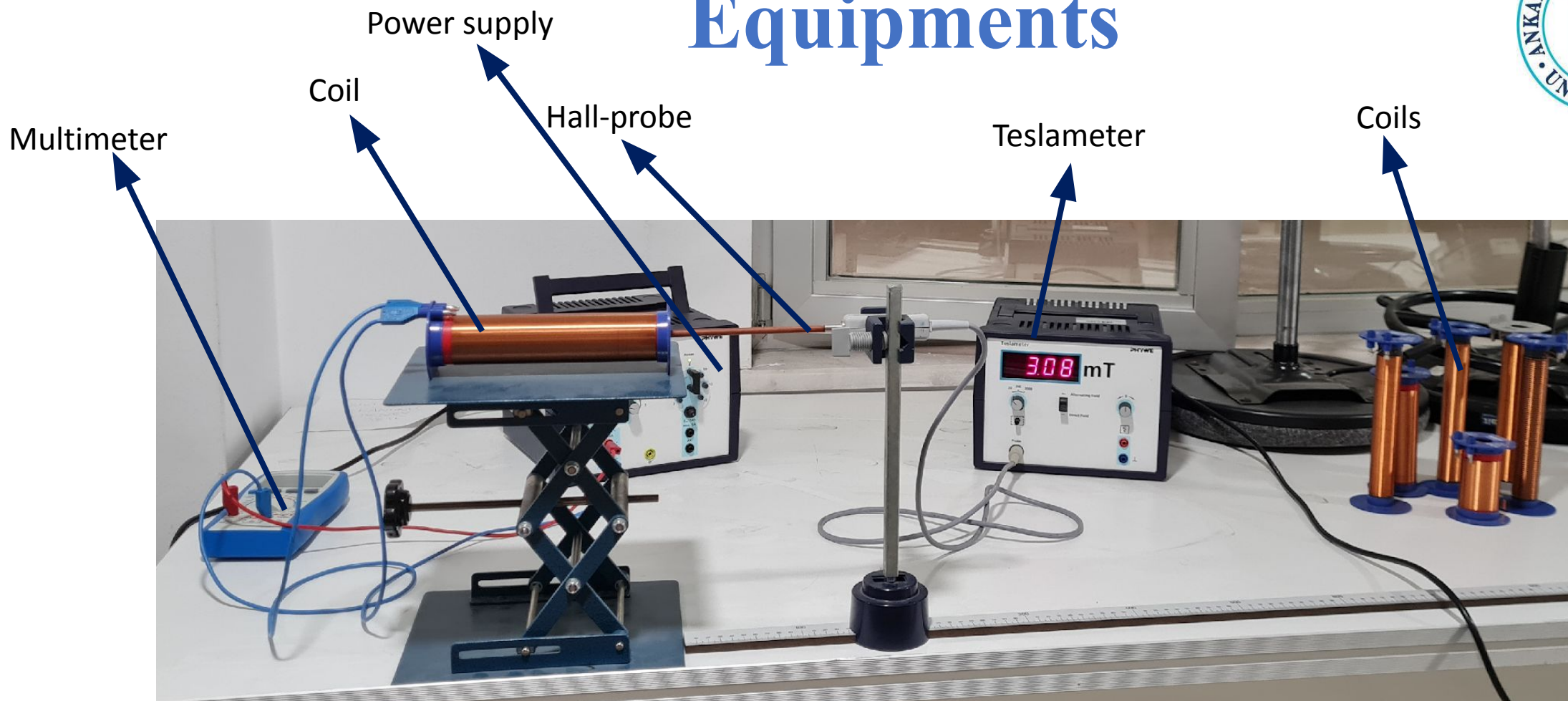
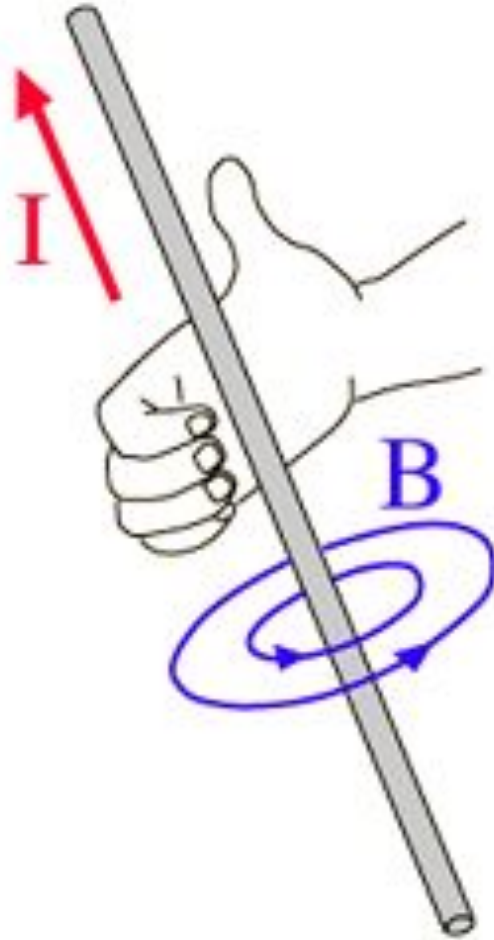


Fig1. Equipments of experiment.

Theory

- To measure the **magnetic flux density** in the **middle of various** wire loops with the Hall probe and to investigate its dependence on the **radius** and number of **turns**.
- To observe changing magnetic flux density along coils.

Theory



B: Magnetic flux density
I: Current

Theory

★ At the centre of the loop ($z = 0$) we obtain

$$\textcircled{1} \quad B(z) = \frac{\mu_0 \cdot I}{2} \cdot \frac{R^2}{(R^2 + z^2)^{3/2}}$$

$$\textcircled{2} \quad B(0) = \frac{\mu_0 \cdot n \cdot I}{2R}$$



B: Magnetic flux density
 μ_0 : Magnetic field constant
I: Current flowing through area
R: Coil Radius
n: Number of turns

Where $\mu_0 = 1.2566 \cdot 10^{-6}$ H/m is the magnetic field constant.

Theory

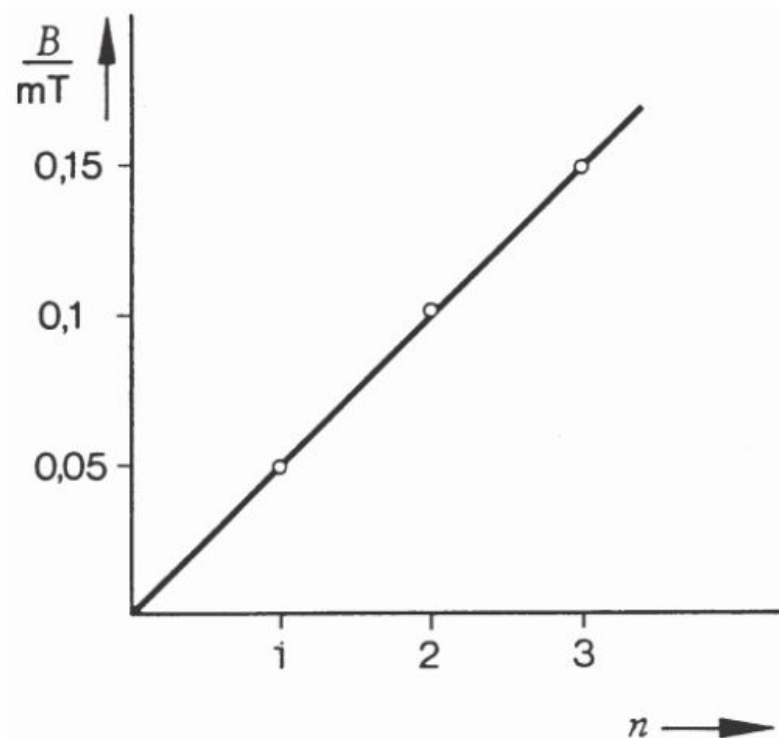


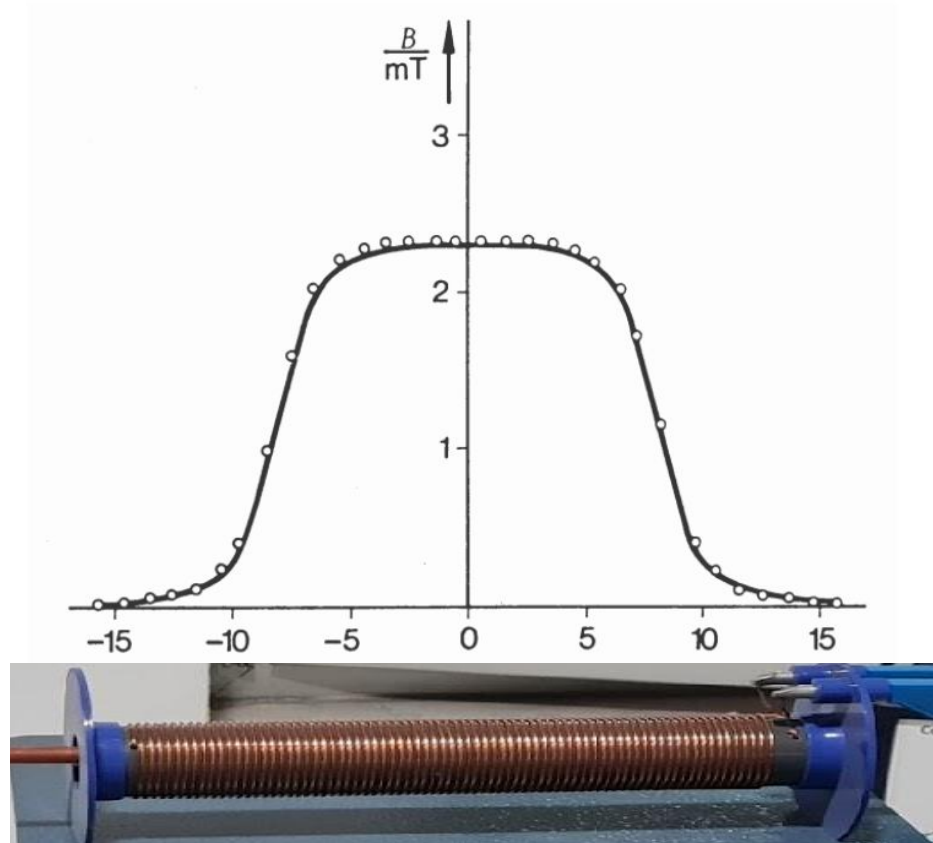
Fig2. Magnetic flux density at the centre of a coil with n turns, as a function of the number of turns (radius 6 cm, current 5 A). (page13)

n	B (mT)
1	0.05
2	0.1
3	0.15

$$2 \quad B(0) = \frac{\mu_0 \cdot n \cdot I}{2R}$$

B: Magnetic flux density
 μ_0 : Magnetic field constant
 I: Current flowing through area
 R: Coil Radius
 n: Number of turns
 l: Length of coil

Experiment



Magnetic flux density decrease the further away from the coil center (page 12)

Fig3. Magnetic flux density along the axis of a coil of length



Report

ATTENTION PLEASE!!!

- You can find your report draft attached file. File name is 'Report Experiment5'
- The calculation must be written (by hand on the paper).
- Name, Surname, Number and department should be written answers sheet.
- If you have a question you can send a mail to **aybuphysicsexperiment5@gmail.com**
- **MEASURED VALUE WRITE IN THE REPORT DRAFT.**

1.) The table was filled after experiment by me.

Calculate $B(0)$. Draw B (measured values)- n curve for current 1A (remarks: see Fig2 and equation 2 from presentation) (40p).

n	l (mm)	R (mm)	$B(0)$ (mT)	
			Measured	Calculated
75	160	13	3.55	
150	160	13	7.00	
300	160	13	14.2	

2.) The table was filled after experiment by me.

Calculate $B(0)$ values for current 1A and discuss the results (60p) (equation 2 from presentation will be used).

n	l (mm)	R (mm)	$B(0)$ (mT)	
			Measured	Calculated
75	160	13	3.55	
150	160	13	7.00	
300	160	13	14.2	
100	53	20	3	
200	105	20	6	
300	160	20	9	
300	160	16	11	